

## GARIBALDI TILLAMOOK, OR, USA

WMO: 997706

Lat: 45.555N

Lon: 123.919W

Elev: 10

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 06-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-----|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |     |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p) |
| 12            | 30.0       | 32.9 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 44.3                  | 50.9 | 40.0 | 51.4 | 7.9                   | 140  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 8             | 8.2                    | 69.8            | N/A  | 66.1 | N/A  | 64.1 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 13.9                 | 0    |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 34.6          | 28.5 | 24.9 | DB                         | 25.1 | 83.5               | 3.7 | 4.0                                                | 22.4 | 86.4       | 20.2 | 88.7       | 18.1 | 90.9       | 15.4 | 93.8 | (4) |
| (5)               |      |      | WB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  | N/A  | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 50.9   | 45.1  | 45.5  | 46.7  | 47.7  | 51.5  | 55.0  | 57.8  | 58.1  | 57.4  | 52.9  | 48.6  | 44.4 |
|                                                                                  | DBStd       | 6.44   | 4.82  | 4.86  | 3.86  | 3.87  | 3.69  | 2.91  | 2.39  | 2.36  | 3.65  | 3.85  | 4.88  | 5.74 |
|                                                                                  | HDD50       | 808    | 163   | 136   | 116   | 86    | 20    | 1     | 0     | 0     | 0     | 15    | 82    | 189  |
|                                                                                  | HDD65       | 5142   | 617   | 545   | 568   | 519   | 419   | 299   | 223   | 214   | 230   | 375   | 493   | 640  |
|                                                                                  | CDD50       | 1144   | 10    | 11    | 13    | 18    | 67    | 152   | 242   | 252   | 222   | 105   | 38    | 14   |
|                                                                                  | CDD65       | 5      | 0     | 0     | 0     | 1     | 1     | 0     | 0     | 1     | 2     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 54     | 0     | 0     | 0     | 7     | 16    | 0     | 7     | 5     | 19    | 0     | 0     | 0    |
|                                                                                  | CDH80       | 10     | 0     | 0     | 0     | 1     | 5     | 0     | 0     | 2     | 2     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 10.4   | 10.3  | 11.0  | 12.9  | 10.8  | 10.0  | 9.3   | 10.5  | 8.5   | 8.6   | 9.6   | 11.9  | 11.9 |
| Precipitation                                                                    | PrecAvg     | 98.9   | 16.1  | 10.7  | 10.0  | 7.9   | 4.8   | 3.4   | 0.9   | 1.6   | 2.8   | 8.3   | 16.7  | 16.3 |
|                                                                                  | PrecMax     | 132.0  | 27.3  | 25.1  | 21.1  | 15.2  | 7.6   | 6.3   | 2.7   | 5.9   | 9.8   | 14.8  | 35.9  | 28.9 |
|                                                                                  | PrecMin     | 73.6   | 7.3   | 2.4   | 1.6   | 3.5   | 1.0   | 1.0   | 0.0   | 0.1   | 0.3   | 1.4   | 6.2   | 7.5  |
|                                                                                  | PrecStd     | 16.9   | 5.0   | 6.1   | 4.1   | 3.2   | 1.9   | 1.4   | 0.8   | 1.4   | 2.2   | 3.7   | 7.3   | 5.6  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 58.4  | 60.1  | 58.8  | 68.7  | 74.1  | 70.0  | 74.3  | 69.4  | 76.8  | 68.6  | 61.1  | 59.9 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | DB     | 55.0  | 56.7  | 55.3  | 59.5  | 63.8  | 63.5  | 66.3  | 66.0  | 69.2  | 63.1  | 58.7  | 56.0 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | DB     | 53.3  | 54.4  | 53.8  | 55.1  | 58.6  | 61.4  | 64.2  | 64.1  | 65.0  | 60.4  | 57.0  | 54.0 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | DB     | 51.9  | 52.7  | 52.7  | 53.4  | 56.3  | 60.0  | 62.7  | 62.6  | 62.8  | 59.1  | 55.5  | 52.3 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.0   | 8.6   | 8.6   | 9.3   | 8.4   | 7.4   | 8.2   | 8.2   | 10.2  | 10.4  | 9.1   | 8.7  |
|                                                                                  |             | MCDBR  | 10.1  | 11.4  | 10.4  | 13.7  | 15.5  | 10.0  | 12.7  | 11.1  | 15.3  | 13.4  | 9.4   | 7.5  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.327  | 0.322 | 0.338 | 0.369 | 0.381 | 0.365 | 0.350 | 0.357 | 0.348 | 0.337 | 0.330 | 0.327 |      |
|                                                                                  | taud        | 2.505  | 2.546 | 2.514 | 2.391 | 2.381 | 2.447 | 2.514 | 2.497 | 2.523 | 2.549 | 2.517 | 2.478 |      |
|                                                                                  | Ebn at Noon | 248    | 275   | 285   | 283   | 281   | 286   | 289   | 283   | 277   | 265   | 245   | 233   |      |
|                                                                                  | Edh at Noon | 22     | 24    | 29    | 36    | 37    | 35    | 32    | 32    | 28    | 24    | 21    | 20    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 378    | 649   | 965   | 1388  | 1718  | 1876  | 2018  | 1732  | 1354  | 810   | 443   | 313   |      |
|                                                                                  | RadStd      | 55     | 81    | 113   | 118   | 153   | 151   | 158   | 130   | 111   | 91    | 56    | 44    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (26 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | +40   |

Nomenclature: See separate page